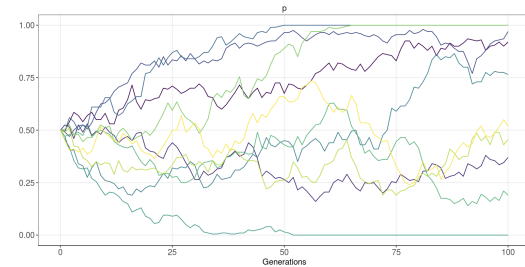
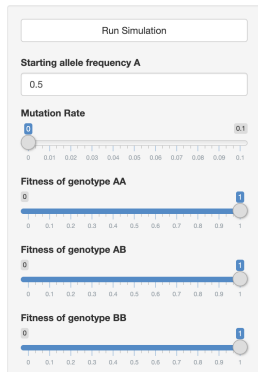


Genetic Drift Simulations

link: <https://cjbattey.shinyapps.io/driftR/>

driftR: Population Genetic Simulations in R



Fixed: 2. Lost: 1.

Final Generation State:

Fis	Hs	Ht	Fst
-0.08	0.23	0.47	0.51

Genetic Drift Activity

- 1) Orient yourself. What are the axes? What variables can you change? What does the plot display when you click “Run Simulation”?
- 2) Set “Starting allele frequency A” to 0.5 (instead of two values, as is default). Set all fitnesses to 1, the Migration Rate to 0, the Number of Populations to 10, and Number of Generations to 100. Make sure you are plotting “p” alone. Run simulations for population sizes of 10, 100, 1000, and 10,000. What pattern do you observe? What does it mean when a colored line hits the top of the graph? What does it mean when a colored line hits the bottom of the graph?
- 3) Repeat the process in 2 but setting the fitness of one of the homozygotes to a value between 0.01 and 0.5. What do you observe?
- 4) Develop a question to address with the driftR.